



How Low Can We Go? Minimizing Interaction Samples for Configurable Systems

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Motivation

T-Wise Interaction Sampling

Test small yet representative list of configurations

- Guarantee coverage criterion (e.g. 2-wise)
- Achieve relatively small sample sizes

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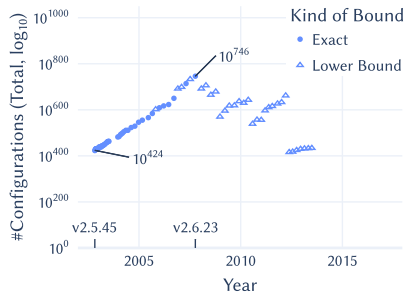
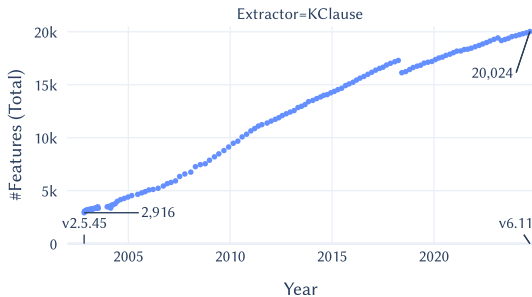
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Problems About Sample Size

Set-Cover is NP-hard

- Absolute quality of solution is unknown
- In general: Optimal sample size is unknown



New Sampling Algorithm: SampLNS

Main Idea of SampLNS

Consider sampling as **set cover** problem

- Cover all *valid interactions* in as few configurations as possible

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Main Tasks of SampLNS

Execute two tasks in parallel:

1. Find and iteratively optimize **upper** bound
2. Find and iteratively optimize **lower** bound

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- Upper bound: Best known sample size
- Lower bound: Measure of absolute quality

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Benefit

- Upper bound: Best known sample size
- Lower bound: Measure of absolute quality
- If lower bound = upper bound
⇒ **Optimal solution found**

Computing a Lower Bound – Mutually Exclusive Interactions

Mutually Exclusive Interactions

Two interactions are mutually exclusive iff there is no valid configuration containing both.

Example: $\{1,2\} \not\subset \{1,-2\}$

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Set of mutually exclusive interactions

$\{1,2\}$

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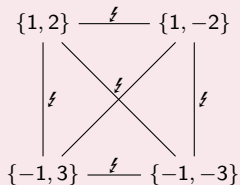
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$$E = \{\{1,2\}, \{1,-2\}, \{-1,3\}, \{-1,-3\}\}$$

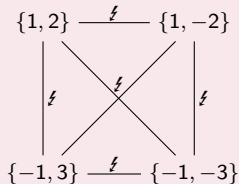
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Benefit

Each interaction in a mutually exclusive set requires a separate configuration.

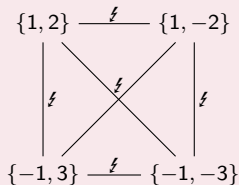
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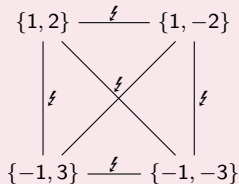
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⇒ **Idea:** Try to find largest set

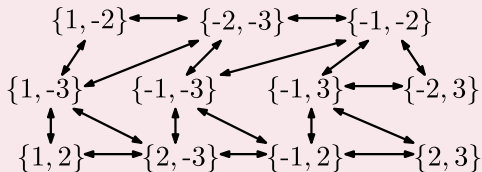
Optimize Lower Bound

LB-LNS

```
 $E = \text{LBSEARCH}()$  [Initial set of mut. excl. interactions]  
while within time limit do  
  Select  $E' \subseteq E$  [The part of the set we remove]  
   $\mathcal{I}' =$  valid interactions that are mut. excl. to  $E \setminus E'$   
   $E'' = \text{OPTLB}(\mathcal{I}')$  [With short time limit]  
   $E = (E \setminus E') \cup E''$  [Combine to new set]  
  Tune  $E'$ -selection based on difficulty of computing  $E''$   
end while  
return  $E$ 
```


Optimize Lower Bound – Example

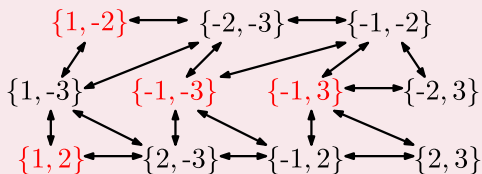
Compatibility Graph



Feature Model: $\mathcal{F} = \{1, 2, 3\}$, $\mathcal{D} = \{\{-1, -3\}\}$

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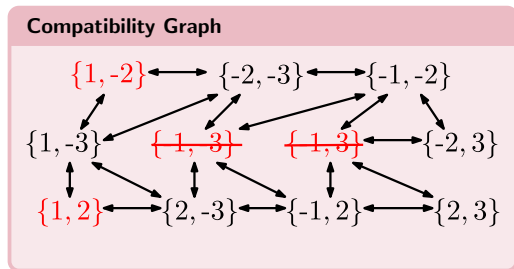


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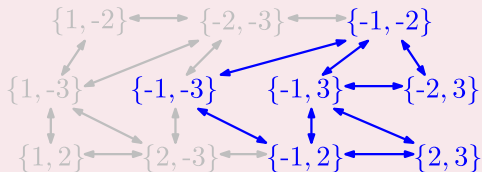


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1. **Initial Set:**
 $E = \{\{1, -2\}, \{1, 2\}, \{-1, -3\}, \{-1, 3\}\}$
2. Remove
 $E' = \{\{-1, -3\}, \{-1, 3\}\}$

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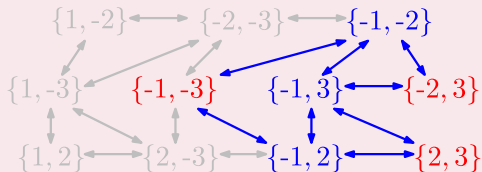
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3. Compute smaller sub problem:

$$\mathcal{I}' = \{\{-1, 2\}, \{-1, -2\}, \{-1, 3\}, \{-1, -3\}, \{2, 3\}, \{-2, 3\}\}$$

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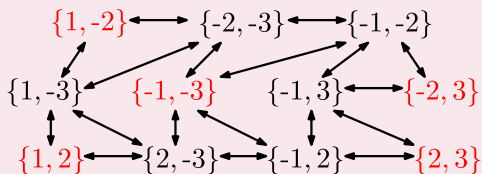
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5. New set:

$$E = (E \setminus E') \cup E''$$

Optimize Upper Bound

SampLNS

S = initial sample computed by YASA ($m = 1$)

Start LB-LNS on a separate thread

while within time limit **and** $|S| > \text{lower bound}$ **do**

 Select $S' \subseteq S$

[The part of the solution we remove]

$\mathcal{I}'$ = interactions covered by S' but not by $S \setminus S'$

$k = |S'|$

[Upper bound on covering $\mathcal{I}'$]

$S'' = \text{OPTSAMPLE}(\mathcal{F}, \mathcal{D}, \mathcal{I}', k)$

[short time limit]

$S = (S \setminus S') \cup S''$

[Combine to new solution]

 Tune S' -selection based on difficulty of computing S''

end while

Terminate LB-LNS and retrieve latest E

return optimized sample S , lower bound $|E|$

Optimize Upper Bound – Example

Feature Model: $\mathcal{F} = \{1, 2, 3, 4\}$, $\mathcal{D} = \{\{1, 2\}, \{3, 4\}\}$

$\Rightarrow$ 22 valid pair-wise interactions:

$$\mathcal{I} = \{ \{3, 4\}, \{1, -3\}, \{2, -4\}, \{1, 3\}, \{-2, 4\}, \{-1, 4\}, \{2, 4\}, \{1, 2\}, \{1, -4\}, \{-2, -3\}, \{-1, -3\}, \{-2, 3\}, \\ \{-1, 3\}, \{3, -4\}, \{-3, 4\}, \{2, -3\}, \{1, -2\}, \{1, 4\}, \{2, 3\}, \{-1, -4\}, \{-2, -4\}, \{-1, 2\} \}$$

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Initial sample: $S = \{\{1, 2, -3, 4\}, \{1, -2, 3, -4\}, \{1, -2, -3, 4\}, \{-1, 2, 3, 4\}, \{-1, 2, 3, -4\}, \{-1, 2, -3, 4\}\}$

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Remove sub-sample: $S' = \{\{1, 2, -3, 4\}, \{-1, 2, 3, 4\}, \{-1, 2, 3, -4\}\}$

$\Rightarrow$ 6 uncovered interactions: $\mathcal{I}' = \{\{1, 2\}, \{2, 3\}, \{3, 4\}, \{2, -4\}, \{-1, 3\}, \{-1, -4\}\}$

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Generate optimal sub-sample:

$$S'' = \text{OPTSAMPLE}(\mathcal{F}, \mathcal{D}, \mathcal{I}', |S'|) = \{\{1, 2, 3, 4\}, \{-1, 2, -3, 4\}\}$$

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Recombine samples:

$$S = (S \setminus S') \cup S'' = \{ \{1, -2, 3, -4\}, \{1, -2, -3, 4\}, \{-1, 2, -3, 4\}, \{1, 2, 3, 4\}, \{-1, 2, -3, 4\} \}.$$

Evaluation Setup

Research Questions

- RQ*₁ Can **existing algorithms** compute samples with a **sample size** equal or **close to the lower bound**?
- RQ*₂ Can **SampLNS** compute samples of **smaller size than existing algorithms**?
- RQ*₃ What is the **computational effort** to **derive lower bounds** and **optimize samples** with SampLNS?
- RQ*₄ Does SampLNS depend on the **initial sample quality**?

Feature Models:

- 46 toy and realistic feature models
- Domains: finance, systems software, e-commerce, gaming, and communication
- Number of features: 9 – 1,408
- Number of constraints: 13 – 15,692

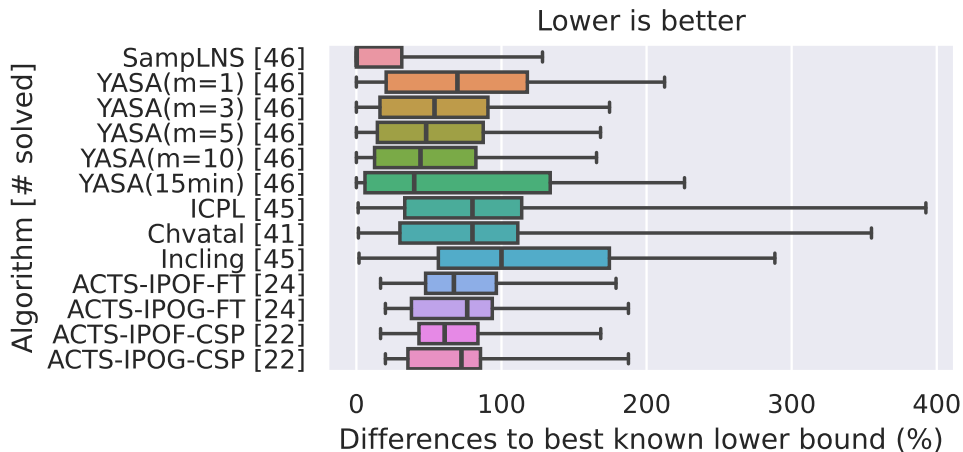
Algorithms:

- Chvátal
- ICPL
- IPOG
- IncLing
- YASA ($m \in \{1, 3, 5, 10\}$)

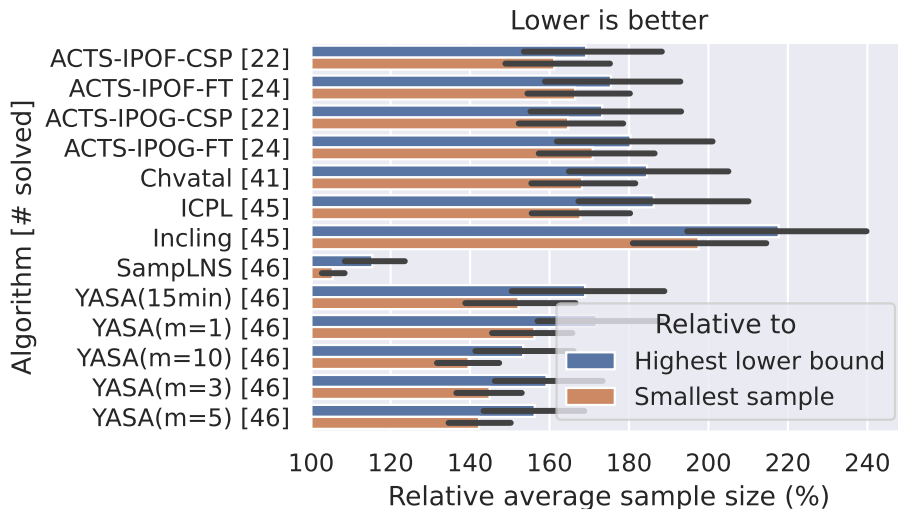
Experiments:

- **Lower Bounds:** Run SampLNS five times with 3 h time limit (180 s per iteration)
- **Comparison:** Run each algorithm five times with 900 s time limit (60 s per iteration)

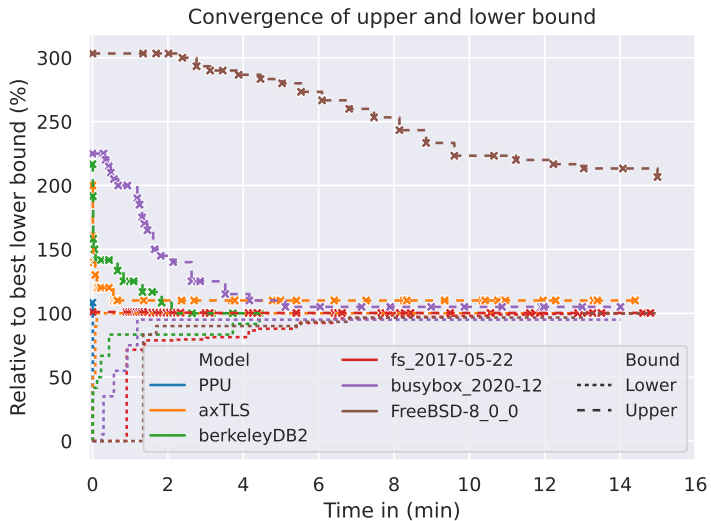
RQ₁: Sample Size – Relative to Lower Bounds



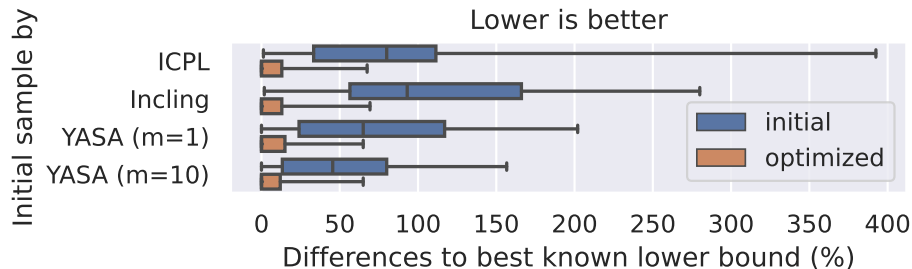
RQ₂: Sample Size – Absolute Average



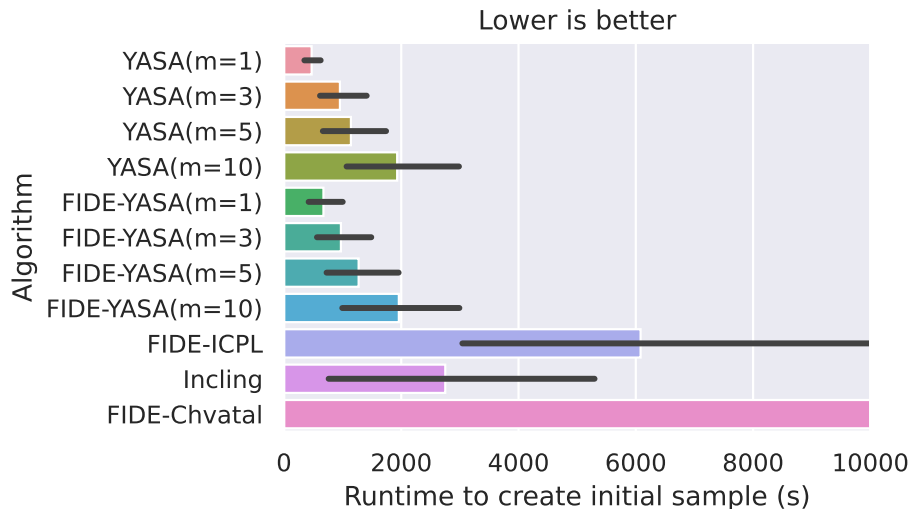
RQ₃: Computational Effort



RQ₄: Importance of Initial Sample



RQ₄: Importance of Initial Sample – Generation Time

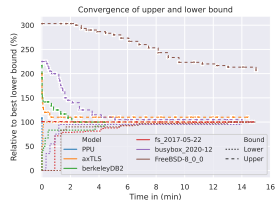
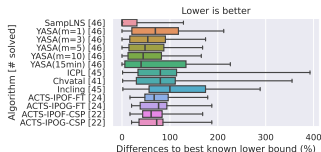
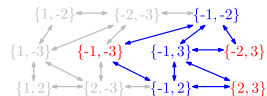


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⇒ New algorithm SampLNS

- Generate samples with (near-)optimality
- Get easily verifiable lower bounds
- Scales for configuration spaces up to 1,400 features



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- Even newer algorithm: **Sammy**

